Chapter 14: Nested and Split-Plot Designs

In a certain experiments, the level of one factor (e.g., factor **B**) are **similar** but **not identical** for different levels of another factor (e.g., factor **A**). For example, if **A** is school and **B** is teacher, teacher 1 will **differ** between the schools.

Such an arrangement is called **nested** or **hierarchical** design, with the levels of factor **B nested under** the level of **factor A**.

This has to be kept in mind when trying to determine if the design is **crossed** or **nested**. To be crossed, the same teacher needs to teach at all the schools.

We will discuss **nested** design and the **split-plot** design in this chapter.

Example: Recall that, in example of 5 of chapter 7, an Engineer was interested in the effects of cutting speed (A), tool geometry (B), and angle (C) on the life (in hours) of a machine tool. Two levels of each factor were chosen and three replicates of a 2^3 factorial design were ran. ABC, AB and BC were confounded in replicate I, II and III, respectively.

Rep 1		Rep 2		Rep 3	
Block 1	**Block 2**	**Block 3**	**Block 4**	**Block 5**	**Block 6**
(1)=22	a =32	(1)=31	a=43	(1)=25	b=50
ab=55	b=35	c=45	b=34	a=29	c=38
ac=40	c=44	ab=47	ac=37	bc=54	ab=46
bc=60	abc=39	abc=41	bc=50	abc=47	ac=36

"Block 1" and "Block 2" meant **different** things within Replicate 1 than within Replicate 2 and Replicate 3.

Blocks were "**nested within replicates**".

Another Example: consider a company that purchases material from three **suppliers** and the material comes in **batches**.

In this case, we might have 4 batches from each supplier, but the batches do **not** have the **same** characteristics of quality when purchased from different suppliers. Therefore, the **batches** are "**nested within suppliers**".

When we have a nested factor and you want to represent this in the model the identity of the batch always requires an **index** of the factor in which it is nested.

Let y_{ijk} denote the response value for k^{th} item of batch j of the supplier i. So, the model is $y_{ijk} = \mu + \tau_i + \beta_{j(i)} + \varepsilon_{(ij)k}$.

The subscript $\boldsymbol{j(i)}$ indicates that $\boldsymbol{j}^{\mathbf{th}}$ level of factor **B** is **nested** under the $\boldsymbol{i^{th}}$ level of factor **A**.

Statistical Model: For $i = 1, 2, \ldots, a, j = 1,2, \ldots, b$ and $k = 1,2, \ldots, n$, let

Response Variable

$$y_{ijk} = \mu + \tau_i + \beta_{j(i)} + \varepsilon_{(ij)k}$$

Overall Mean

Supplier Effect

Batch Effect

Random Error

Note: it is useful to think of replicates as being nested under the treatment combinations; thus, $(ij)k$ is used for the error term.

Because **not** every level of B appears with every level of A, **interaction** makes **no** sense here.

A and B can be fixed and/or random. Since B is **nested** within A, factor **B** is **random** whenever **A** is **random**.

Statistical Model: For $i = 1, 2, \ldots, a, j = 1,2, \ldots, b$ and $k = 1,2, \ldots, n$, let

$$y_{ijk} = \mu + \tau_i + \beta_{j(i)} + \varepsilon_{(ij)k}$$

Define

$$SS_{Total} = \sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{n}(y_{ijk} - \bar{y}_{...})^2 ,$$

$$SS_A = \sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{n}(\bar{y}_{i..} - \bar{y}_{...})^2 ,$$

$$SS_{B(A)} = \sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{n}(\bar{y}_{ij.} - \bar{y}_{i..})^2 ,$$

$$SS_{Error} = \sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{n}(y_{ijk} - \bar{y}_{ij.})^2$$

It can be shown that

$$SS_{Total} = SS_A + SS_{B(A)} + SS_{Error}$$

Statistical Model: For $i = 1, 2, \ldots, a, j = 1,2, \ldots, b$ and $k = 1,2, \ldots, n$, let

$$y_{ijk} = \mu + \tau_i + \beta_{j(i)} + \varepsilon_{(ij)k}$$

Note:

- There are $\boldsymbol{abn - 1}$ degrees of freedom for $\boldsymbol{SS_{Total}}$,
- There are $\boldsymbol{a - 1}$ degrees of freedom for $\boldsymbol{SS_A}$,
- There are $\boldsymbol{a(b - 1)}$ degrees of freedom for $\boldsymbol{SS_{B(A)}}$, and
- There are $\boldsymbol{ab(n - 1)}$ degrees of freedom for **error**.
- In a "**two-stage**" nested design we might have **both** factors are **fixed**. Then, we assume

$$\sum_{i=1}^{a} \tau_i = 0 \; and \sum_{j=1}^{b} \beta_{j(i)} = 0$$

- We might have **A** is **fixed** and **B** is **random** or **both** A and B are **random**. In this case, we substitute the above condition by

$$\tau_i \sim N(0, \sigma_\tau^2) \text{ and/or } \beta_{j(i)} \sim N(0, \sigma_\beta^2)$$

The **ANOVA table** when **both** A and B are **fixed**.

Source	df	SS	MS	F
A	$\mathrm{a}-1$	SS_A	$MS_A = \frac{SS_A}{\mathrm{a}-1}$	$\mathrm{F} = \frac{MS_A}{MS_E}$
B(A)	$\mathrm{a(b}-1)$	$SS_{B(A)}$	$MS_{B(A)} = \frac{SS_{B(A)}}{a(b-1)}$	$\mathrm{F} = \frac{MS_{B(A)}}{MS_E}$
Error	$ab(n-1)$	SS_E	$MS_E = \frac{SS_E}{ab(n-1)}$	
Total	$\mathrm{abn}-1$	SST		

Example 1: The surface finish of metal parts made on **four** machines is being studied.

Different operators are used on each machine. Each machine is run by **three** different operators and **two** specimens from each operator are tested. The data are given in the following table.

Machine 1			Machine 2			Machine 3			Machine 4		
1	2	3	1	2	3	1	2	3	1	2	3
79	94	46	92	85	76	88	53	46	36	40	62
62	74	57	99	79	68	75	56	57	53	56	46

```
> y = c(79, 94, 46, 92, 85, 76, 88, 53, 46, 36, 40, 62,
        62, 74, 57, 99, 79, 68, 75, 56, 57, 53, 56, 47)
```

```
> Machine = as.factor(rep(1:4, each=3, times = 2))
> Operator = as.factor(rep(1:3, times=8))
```

R Codes to Obtain ANOVA Table:

```
> model =  lm(y~machine + operator %in% machine)
> anova(model)
```

```
Analysis of Variance Table

Response: y
                 Df Sum Sq Mean Sq F value   Pr(>F)
machine           3 3617.7 1205.89 14.2709 0.000291 ***
machine:operator  8 2817.7  352.21  4.1681 0.013408 *
Residuals        12 1014.0   84.50
```

- The **appropriate test statistic** for testing the effect of factors A and B depend on whether A and B are **fixed** or **random**.

- When **B** is a **random** factor **nested** in **A**, we think of it as the **replicates** for A. So whether factor A is a fixed or random factor the error term for testing the hypothesis about A is based on the mean squares due to B(A) which is read "B nested in A".

Statistical Model: For $i = 1, 2, \ldots, a, j = 1,2, \ldots, b$ and $k = 1,2, \ldots, n$, let

$$y_{ijk} = \mu + \tau_i + \beta_{j(i)} + \varepsilon_{(ij)k}$$

Note:

E(MS)	A is fixed B is fixed	A is fixed B is random	A is random B is random
$E(MS_A)$	$\sigma^2 + \frac{bn\sum\tau_i^2}{a-1}$	$\sigma^2 + n\,\sigma_\beta^2 + \frac{bn\sum\tau_i^2}{a-1}$	$\sigma^2 + n\,\sigma_\beta^2 + bn\sigma_\tau^2$
$E(MS_{B(A)})$	$\sigma^2 + \frac{n\sum\beta_{j(i)}^2}{a(b-1)}$	$\sigma^2 + n\,\sigma_\beta^2$	$\sigma^2 + n\,\sigma_\beta^2$
$E(MS_{Error})$	σ^2	σ^2	σ^2

When **B** is a **random** factor **nested** in **A**, we think of it as the **replicates** for A. So whether factor A is a fixed or random factor the error term for testing the hypothesis about A is based on the mean squares due to B(A) which is read "B nested in A".

Example 1 (Cont.): Suppose machine (factor A) were randomly chosen. Therefore, both factors were random.

```
Analysis of Variance Table

Response: y
                 Df Sum Sq Mean Sq F value   Pr(>F)
machine           3 3617.7 1205.89 14.2709 0.000291 ***
machine:operator  8 2817.7  352.21  4.1681 0.013408 *
Residuals        12 1014.0   84.50
```

The appropriate test statistic for H_0: $\sigma_\tau^2 = 0$ is

$$F_O = \frac{MS_A}{MS_{B(A)}} = \frac{1205.89}{352.21} = 3.424 \rightarrow p - value = P(F_8^3 > 3.424) = 0.0728$$

The appropriate test statistic for testing H_0: $\sigma_\beta^2 = 0$ is

$$F_O = \frac{MS_{B(A)}}{MS_{Error}} = \frac{352.21}{84.50} = 4.1681 \rightarrow p - value = P\left(F_{12}^8 > 4.168\right) = 0.0134$$

R Codes to Retrieve the F-values:

```
> model_anova = anova(model)
> ms = as.vector(model_anova[[3]])
> df = as.vector(model_anova[[1]])
> F0=c(ms[1]/ms[2],ms[2]/ms[3]

> cat("F to test effect of machines is", F0[1], "and p-value is",
1-pf(F0[1],df[1],df[2]), "\n")
```

F to test effect of machines is 3.423794 and p-value is 0.07279676

```
> cat("F to test effect of operators(machines) is", F0[2], "and p-value
      is",1-pf(F0[2],df[2],df[3]), "\n")
```

F to test effect of operators(machines) is 4.168146 and p-value is 0.01340834

Example 1 (Cont.): Suppose machine (factor A) were randomly chosen. Therefore, both factors were random.

```
Analysis of Variance Table

Response: y
                 Df Sum Sq Mean Sq F value   Pr(>F)
machine           3 3617.7 1205.89 14.2709 0.000291 ***
machine:operator  8 2817.7  352.21  4.1681 0.013408 *
Residuals        12 1014.0   84.50
```

The moment estimate of variance components are:

$$\hat{\sigma}^2 = MS_{Error} = 84.50,$$

$$\hat{\sigma}_\beta^2 = \frac{MS_{B(A)} - MS_{Error}}{n} = \frac{352.21 - 84.50}{2} = 133.86$$

$$\hat{\sigma}_\tau^2 = \frac{MS_A - MS_{B(A)}}{bn} = \frac{1205.89 - 352.21}{3(2)} = 142.28$$

R Codes to Obtain the Moment Estimates of Variance Components:

```
>n = 2 ; a = 4; b = 3
> sigma2_tau = (ms[1]-ms[2])/(n*b)
> sigma2_beta = (ms[2]-ms[3])/n

> cat("Method of Moments Variance Component Estimates","\n",
   "Variance of Machine = ", sigma2_tau, "\n",
   "Variance of Operator = ", sigma2_beta, "\n",
   "Variance of Error = ", ms[3], "\n")
```

Method of Moments Variance Component Estimates
Variance of Machine = 142.2801
Variance of Operator = 133.8542
Variance of Error = 84.5

R Codes to REML Estimates:

```
> library(lme4)
> model2 = lmer(y~1+(1|Machine/Operator))
> summary(model2)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: y ~ 1 + (1 | Machine/Operator)

REML criterion at convergence: 189.9

Scaled residuals:
     Min       1Q   Median       3Q      Max
-1.17044 -0.69972  0.00112  0.68815  1.50964

Random effects:
 Groups           Name        Variance Std.Dev.
 Operator:Machine (Intercept) 133.9    11.570
 Machine          (Intercept) 142.3    11.928
 Residual                      84.5     9.192
Number of obs: 24, groups:  Operator:Machine, 12; Machine, 4

Fixed effects:
            Estimate Std. Error t value
(Intercept)   65.833      7.088   9.287
```

The results can be easily extended to the case of m completely nested factors. Such a design would be called an ***m*-stage nested** design.

Consider a design with three factors A, B and C where **B** is **nested in A** and **C** is **nested in B**.

For $i = 1,2, \dots a, j = 1,2, \dots b, \quad k = 1,2, \dots, c$ and $l = 1,2, \dots n$, let

$$y_{ijkl} = \mu + \tau_i + \beta_{j(i)} + \gamma_{k(ij)} + \varepsilon_{(ijk)l}$$

Define

$$SS_{Total} = \sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{c}\sum_{l=1}^{n}(y_{ijkl} - \bar{y}_{....})^2 \,, SS_A = \sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{c}\sum_{l=1}^{n}(\bar{y}_{i...} - \bar{y}_{....})^2$$

$$SS_{B(A)} = \sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{c}\sum_{l=1}^{n}(\bar{y}_{ij..} - \bar{y}_{i...})^2 \; SS_{C(B)} = \sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{c}\sum_{l=1}^{n}(\bar{y}_{ijk.} - \bar{y}_{ij..})^2$$

It can be shown that

$$SS_{Total} = SS_A + SS_{B(A)} + SS_{C(B)} + SS_{Error}$$

The **ANOVA table** when **all** A and B a d C are **fixed**.

Source	df	SS	MS	F
A	$\mathrm{a}-1$	SS_A	$MS_A = \frac{SS_A}{\mathrm{a}-1}$	$\mathrm{F} = \frac{MS_A}{MS_E}$
B(A)	$\mathrm{a}(\mathrm{b}-1)$	$SS_{B(A)}$	$MS_{B(A)} = \frac{SS_{B(A)}}{a(b-1)}$	$\mathrm{F} = \frac{MS_{B(A)}}{MS_E}$
C(B)	$ab(c-1)$	$SS_{C(B)}$	$MS_{C(B)} = \frac{SS_{C(B)}}{ab(c-1)}$	$\mathrm{F} = \frac{MS_{C(B)}}{MS_E}$
Error	$abc(n-1)$	SS_E	$MS_E = \frac{SS_E}{abc(n-1)}$	
Total	$\mathrm{abcn}-1$	SST		

Note that if **A** and **B** are **fixed** and **C** is **random**, then

$$E(MS_A) = \sigma^2 + n\,\sigma_\gamma^2 + \frac{bcn\sum \tau_i^2}{a-1}$$

$$E\left(MS_{B(A)}\right) = \sigma^2 + n\,\sigma_\gamma^2 + \frac{cn\sum \beta_{j(i)}^2}{a(b-1)}$$

$$E\left(MS_{C(B)}\right) = \sigma^2 + n\,\sigma_\gamma^2 \quad \text{and} \quad E(MS_{Eror}) = \sigma^2$$

Therefore, the appropriate test statistic for testing $H_0: \tau_i = 0$ is $F = \frac{MS_A}{MS_{C(B)}}$.

The appropriate test statistic for testing $H_0: \beta_{j(i)} = 0$ is $F = \frac{MS_{B(A)}}{MS_{C(B)}}$

The appropriate test statistic for testing $H_0: \sigma_\gamma^2 = 0$ is $F = \frac{MS_{C(B)}}{MS_{Error}}$.

To fit the model use **lm(y ~ A + B%inA% + C%in%B)** in R.

Design With Both Nested and Factorial Factors: Occasionally in a multifactor experiment, some factors are arranged in a factorial layout and other are nested. These design sometimes are called **nested-factorial design**.

Example 2 (Example 14.2 of textbook): An industrial engineer is studying the hand insertion of electronic components on printed circuit boards to improve the speed of the assembly operation.

He has designed **three assembly fixtures** and **two workplace layout** that seem promising.

Operators are required to perform the assembly, and it is decided to randomly select **four operators** for each **fixture-layout combinations**.

However because the workplaces are in different locations within the plant, it is difficult to use the same four operators for each layout. Therefore, the **four operators** chosen for **layout 1** are **different** individuals from the **four operators** chosen from **layout 2**.

Example 2 (Cont.): Because there are only **three fixtures** and **two layouts**, but the **operators** are chosen at **random**, this is a **mixed model**.

The treatment combinations in this design are run in random order, and **two replicates** are obtained. .

The assembly times are measured in seconds and are shown in the following table.

	Layout 1				Layout 2			
Operator	1	2	3	4	1	2	3	4
Fixture 1	22	23	28	25	26	27	28	24
	24	24	29	23	28	25	25	23
Fixture 2	30	29	30	27	29	30	24	28
	27	28	32	25	28	27	23	30
Fixture 3	25	24	27	26	27	26	24	28
	21	22	25	23	25	24	27	27

Example 2 (Cont.): In this experiment, **operators** are **nested** within the levels of **layouts**, whereas **fixtures** and **layouts** are arranged in a **factorial**.

Statistical Model: For $i = 1, 2, 3, j = 1,2, k = 1,2,3,4$, and $l = 1,2$ let

Assemby Time

The Effect of jth Layout

Fixture × Layout Interaction

Error Term

$$y_{ijkl} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \gamma_{k(j)} + (\tau\gamma)_{ik(j)} + \varepsilon_{(ijk)l}$$

The Effect of ith Fixture

The Effect of kth Operator Within the jth Layout

Fixture × Operators Within Layout Interaction

Note that if **A = Fixture** and **B = Layout** are **fixed** and **C = Operator** is **random**, then with $a = 3, b = 2, c = 4$ and $n = 2$, we have

$$E(MS_A) = \sigma^2 + n\,\sigma^2_{(\tau\gamma)} + \frac{bcn\sum\tau_i^2}{a-1} = \sigma^2 + 2\,\sigma^2_{(\tau\gamma)} + 8\sum\tau_i^2$$

$$E(MS_B) = \sigma^2 + na\,\sigma^2_{\gamma} + \frac{acn\sum\beta_j^2}{b-1} = \sigma^2 + 6\,\sigma^2_{\gamma} + 24\sum\beta_j^2$$

$$E(MS_{AB}) = \sigma^2 + n\,\sigma^2_{(\tau\gamma)} + \frac{cn\sum\sum(\tau\beta)_{ij}^2}{(a-1)(b-1)} = \sigma^2 + 2\,\sigma^2_{(\tau\gamma)} + 4\sum\sum(\tau\beta)_{ij}^2$$

$$E\left(MS_{C(B)}\right) = \sigma^2 + na\sigma^2_{\gamma} = \sigma^2 + 6\sigma^2_{\gamma}$$

$$E\left(MS_{AC(B)}\right) = \sigma^2 + n\,\sigma^2_{(\tau\gamma)} = \sigma^2 + 2\,\sigma^2_{(\tau\gamma)}$$

and $\quad E(MS_{Eror}) = \sigma^2$

Note that if **A = Fixture** and **B = Layout** are **fixed** and **C = Operator** is **random**, therefore, the appropriate test statistic for testing $H_0: \tau_i = 0$ is

$$F = \frac{MS_A}{MS_{AC(B)}}.$$

The appropriate test statistic for testing $H_0: \beta_j = 0$ is

$$F = \frac{MS_{B(A)}}{MS_{C(B)}}$$

The appropriate test statistic for testing $H_0: (\tau\beta)_{ij} = 0$ is

$$F = \frac{MS_{AB}}{MS_{AC(B)}}$$

The appropriate test statistic for testing $H_0: \sigma_\gamma^2 = 0$ is

$$F = \frac{MS_{C(B)}}{MS_{Error}}.$$

The appropriate test statistic for testing $H_0: \sigma_{(\tau\gamma)}^2 = 0$ is

$$F = \frac{MS_{AC(B)}}{MS_{Error}}.$$

R Codes to Enter Data:

```
> Time = c(22, 23, 28, 25, 26, 27, 28, 24,
           24, 24, 29, 23, 28, 25, 25, 23,
           30, 29, 30, 27, 29, 30, 24, 28,
           27, 28, 32, 25, 28, 27, 23, 30,
           25, 24, 27, 26, 27, 26, 24, 28,
           21, 22, 25, 23, 25, 24, 27, 27)
```

	Time	Fixture	Layout	Operator
1	22	1	1	1
2	23	1	1	2
3	28	1	1	3
4	25	1	1	4
5	26	1	2	1
6	27	1	2	2
7	28	1	2	3
8	24	1	2	4
9	24	1	1	1
10	24	1	1	2

```
> Fixture  = as.factor(rep(1:3, each=16))
> Layout   = as.factor(rep(1:2, each=4, times = 6))
> Operator = as.factor(rep(1:4, times=12))

> data <- data.frame(Time, Fixture, Layout, Operatour)
> head(data,10)
```

R Codes to Obtain ANOVA Table:

```
> model =  lm(Time~(Fixture + Layout)^2 +
                    (Operator + Fixture*Operator)%in%Layout)

> anova(model)
Analysis of Variance Table

Response: Time
                      Df Sum Sq Mean Sq F value    Pr(>F)
Fixture                2 82.792  41.396 17.7411 1.862e-05 ***
Layout                 1  4.083   4.083  1.7500  0.198344
Fixture:Layout         2 19.042   9.521  4.0804  0.029828 *
Layout:Operator        6 71.917  11.986  5.1369  0.001606 **
Fixture:Layout:Operator 12 65.833   5.486  2.3512  0.036043 *
Residuals             24 56.000   2.333
```

R Codes to Retrieve the Mean Squares and Degrees of Freedoms:

```
> model_anova = anova(model)

> ms = as.vector(model_anova[[3]])

> MS_A <- ms[1];      MS_B <- ms[2];      MS_AB <- ms[3]
> MS_CinB <- ms[4]; MS_ACinB <- ms[5] ; MSE <- ms[6]

> df = as.vector(model_anova[[1]])

> df_A <- df[1];      df_B <- df[2];      df_AB <- df[3]
> df_CinB <- df[4]; df_ACinB <- df[5]; df_E <- df[6]
```

R Codes to Retrieve the F-Tests:

```
>F_A <- MS_A/MS_ACinB
> F_B <- MS_B/MS_CinB
> F_AB <- MS_AB/MS_ACinB
> F_C <- MS_CinB/MSE
> F_ACinB <- MS_ACinB/MSE

> cat("F to test effect of fixtures (A) is", F_A, "and p-value is",
        1-pf(F_A, df_A, df_ACinB), "\n")
```

F to test effect of fixtures (A) is 7.54557 and p-value is 0.007553076

R Codes to Retrieve the F-Tests:

```
> cat("F to test effect of layouts (B) is", F_B, "and p-value is",
        1-pf(F_B, df_B, df_CinB), "\n")
```

F to test effect of layouts (B) is 0.3406721 and p-value is 0.5807041

```
> cat("F to test effect of fixture/layout interaction (AB) is", F_AB, "and
        p-value is", 1-pf(F_AB, df_AB, df_ACinB), "\n")
```

F to test effect of fixture/layout interaction (AB) is 1.735443 and p-value is 0.2177691

R Codes to Retrieve the F-Tests:

```
> cat("F to test effect of operators (CinB) is", F_C, "and p-value is",
        1-pf(F_C, df_CinB, df_E), "\n")
```

F to test effect of operators (CinB) is 5.136905 and p-value is 0.001605804

```
> cat("F to test effect of fixture/operator (in layouts) interaction (AC in
        B) is", F_ACinB, "and p-value is", 1-pf(F_ACinB, df_ACinB,
        df_E), "\n")
```

F to test effect of fixture/operator (in layouts) interaction (AC in B) is 2.35119 and p-value is 0.03604336

The results can be summarized in the following table:

Factor	F-test	P-value
Fixture	7.546	0.0076
Layout	0.341	0.5807
Fixture×Layout	1.736	0.2178
Operator(Layout)	5.138	0.0016
Operator×Fixture(Layout)	2.351	0.0360

- **Fixtures** are significantly **different**.

- **Layout** does **not** have a significant effect on assembly time.
- **Fixture ×Layout** does **not** have significant effect on assembly time.

- **Operators differ** significantly **within** at least one of the **layouts**.
- **Fixture × Operator interaction** is **significant within** at least one of the **layouts** (so some operators are quicker than others, using the same fixtures)

Fixtures are significantly **different**.

Tukey's confidence intervals for $\tau_i - \tau_{i'}$ for $i \neq i' = 1,2,3$ are:

$$\bar{y}_{i\ldots} - y_{i'\ldots} \pm \frac{qtukey\left(1-\alpha, a, df\left(MS_{AC(B)}\right)\right)}{\sqrt{2}} \sqrt{\frac{2}{r} MS_{AC(B)}}$$

For $\alpha = 0.05, a = 3,$ and $df\left(MS_{AC(B)}\right)$*=12, qtukey(0.95,3,12) is 3.772929. With* $r = nbc = 16$, number of observation used in a each treatment mean, we have

$$\frac{qtukey\left(1-\alpha, a, df\left(MS_{AC(B)}\right)\right)}{\sqrt{2}} \sqrt{\frac{2}{r} MS_{AC(B)}} = \frac{3.772929}{\sqrt{2}} \sqrt{\frac{2}{16}(5.486)}$$
$$= 2.209$$

Any mean difference larger than 2.209 in absolute value is significant.

Note that: $\bar{y}_{1\ldots} = 25.2500, \bar{y}_{2\ldots} = 27.9375$ and $\bar{y}_{3\ldots} = 25.0625$. Therefore,

$$|\bar{y}_{1\ldots} - \bar{y}_{2\ldots}| = |25.2500 - 27.9375| = 2.6875 > \mathrm{ME} = 2.209$$
$$|\bar{y}_{1\ldots} - \bar{y}_{3\ldots}| = |25.2500 - 25.0625| = 0.1875 < \mathrm{ME} = 2.209$$
$$|\bar{y}_{2\ldots} - \bar{y}_{3\ldots}| = |27.9375 - 25.0625| = 2.8750 > \mathrm{ME} = 2.209$$

$\mu_3 \qquad \mu_1 \qquad \mu_2$

See R codes on Eclass to calculate margin of error and averages.

The estimates of variance components can be found by using moment method or REML method.

Note that if **A = Fixture** and **B = Layout** are **fixed** and **C = Operator** is **random**, then with $a = 3, b = 2, c = 4$ and $n = 2$, we have

$$E\left(MS_{C(B)}\right) = \sigma^2 + na\sigma_\gamma^2 = \sigma^2 + 6\sigma_\gamma^2$$

$$E\left(MS_{AC(B)}\right) = \sigma^2 + n\,\sigma_{(\tau\gamma)}^2 = \sigma^2 + 2\,\sigma_{(\tau\gamma)}^2$$

$$\text{and} \quad E(MS_{Eror}) = \sigma^2$$

Therefore, the moment estimates of variance components are:

$$\hat{\sigma}^2 = MS_{Eror} = 2.333$$

$$\hat{\sigma}_\gamma^2 = \frac{MS_{C(B)} - MS_{Eror}}{na} = \frac{11.986 - 2.333}{6} = 1.609$$

$$\hat{\sigma}_{(\tau\gamma)}^2 = \frac{MS_{AC(B)} - MS_{Eror}}{n} = \frac{5.486 - 2.333}{2} = 1.577$$

R Codes to Obtain Moment Estimates of Variance Components

```
> n=2; a=3
> sigma2_gamma = (MS_CinB-MSE)/(n*a)
> sigma2_taugamma = (MS_ACinB-MSE)/n
>cat("Method of Moments Variance Component Estimates","\n",
   "Estimate of gamma variance (operator in layout) is",
    sigma2_gamma , "\n", "Estimate of tau*gamma variance (fixture x
     operator in layout) is", sigma2_taugamma, "\n",
    "Estimate of error variance  is", MSE, "\n")
```

Method of Moments Variance Component Estimates

Estimate of gamma variance (operator in layout) is 1.608796

Estimate of tau*gamma variance (fixture x operator in layout) is 1.576389

Estimate of error variance is 2.333333

R Codes to Obtain REML Estimates of Variance Components

```
> library(lme4)
> model2 = lmer(Time~1+ Fixture + Layout + Fixture*Layout +
                (1|Layout:Operator) + (1|Layout:Fixture:Operator))
```

```
Linear mixed model fit by REML ['lmerMod']
Formula:
Time ~ 1 + Fixture + Layout + Fixture * Layout + (1 | Layout:Operator) +
    (1 | Layout:Fixture:Operator)

REML criterion at convergence: 187.3

Scaled residuals:
     Min       1Q   Median       3Q      Max
-1.47784 -0.69541 -0.04367  0.70205  1.37326

Random effects:
 Groups                   Name        Variance Std.Dev.
 Layout:Fixture:Operator (Intercept) 1.576    1.256
 Layout:Operator         (Intercept) 1.083    1.041
 Residual                            2.333    1.528
Number of obs: 48, groups:  Layout:Fixture:Operator, 24; Layout:Operator, 8

Fixed effects:
                 Estimate Std. Error t value
(Intercept)       24.7500     0.9781  25.305
Fixture2           3.7500     1.1711   3.202
Fixture3          -0.6250     1.1711  -0.534
Layout2            1.0000     1.3832   0.723
Fixture2:Layout2  -2.1250     1.6562  -1.283
Fixture3:Layout2   0.8750     1.6562   0.528

Correlation of Fixed Effects:
             (Intr) Fixtr2 Fixtr3 Layot2 Fx2:L2
Fixture2     -0.599
Fixture3     -0.599  0.500
Layout2      -0.707  0.423  0.423
Fixtr2:Lyt2   0.423 -0.707 -0.354 -0.599
Fixtr3:Lyt2   0.423 -0.354 -0.707 -0.599  0.500
```

The results can be summarized in the following table:

Factor	F-test	P-value
Fixture	7.546	0.0076
Layout	0.341	0.5807
Fixture×Layout	1.736	0.2178
Operator(Layout)	5.138	0.0016
Operator×Fixture(Layout)	2.351	0.0360

R Codes to Obtain Correct ANOVA Table Using REML Estimates:

> anova(model2)

```
Analysis of Variance Table
               Df Sum Sq Mean Sq F value
Fixture         2 35.213 17.6063  7.5456
Layout          1  0.795  0.7949  0.3407
Fixture:Layout  2  8.099  4.0494  1.7354
```

Note that **anova(model)** gives F-values only for fixed effect.

Split-plot Design: In some multifactor factorial experiments, we may be **unable** to **completely randomize** the order of the runs.

Sometimes the levels of one or more factors in the design are **more difficult** to **change** or **require more experimental material** to evaluate. In this situation, complete randomization of factor-level combinations to experimental units could make the experiment **much more time consuming** or perhaps **impossible** to conduct.

This often results in a generalization of the factorial design called a **split-plot design**. In a split-plot design, the randomization is such that the hard-to-vary factors are not changed as frequently as easy-to-vary factors.

The name split-plot comes from the original application in agricultural experiments, where the levels of some factors (called whole-plot factors) could only be varied between plots of land, while the levels of other factors could be varied within a plot.

Example 3: A factorial experiment was devised to determine if there was a way to change the process of making the orange cookies that would reduce the spreading during baking.

The factors that were chosen to be varied were:

- A: The amount of shortening in the dough batch (80% of what the recipe called for or 100%)
- B: The baking temperature (below, at, or above the temperature called for the recipe)
- C: the temperature of the cookies sheet upon which the cookies were placed to be baked (hot out of the oven, or cooled to room temperature)

A response that could quantify the objective of the experiment was the diameter of the backed cookies.

Example 3 (Cont.): The following table shows a completely randomized list for this 3×2^2 factorial experiment.

Batch	Shortening	Bake Temperature	Tray Temperature
1	100%	Low	Room T
2	100%	Low	Hot
3	100%	Norm	Room T
4	100%	High	Room T
5	80%	High	Room T
6	80%	Norm	Room T
7	100%	High	Hot
8	100%	Norm	Hot
9	80%	Low	Room T
10	80%	High	Hot
11	80%	Norm	Hot
12	80%	Low	Hot

Note that the experimental unit is a batch of cookies and this plan calls for backing 12 batches.

There are no replicate batches, and therefore, no estimate of the variance experimental error from replicates in this plan.

Example 3 (Cont.): The cookie-making process consist of the following two-steps:

1) Mix cookie dough batch
2) Bake cookies

Note:

- Factor **A** was related to process **step 1**, while factor s **B** and **C** were related to process **step 2**.

- The **amount of shortening** was a **hard-to–vary** factor because each time it was changed it required making a new batch of cookie dough, while the baking temperature and tray temperature were easy to vary.

- Once a batch of dough was made, there was enough to make six tray of cookies, and all six combinations of baking temperature and tray temperature could be tested within each dough batch, greatly reducing the total number of batches required.

Example 3 (Cont.): An alternative way of designing the experiment would be to follow a two-step plan.

First, plan to make four batches of cookie dough and randomly assign two batches to use 80% of the recipe recommended amount of shortening and two batches to receive the full amount of shortening recommended by the recipe.

This represent a completely randomized design in one factor and the experimental unit is a batch of cookie dough. This first step is called the **whole-plot design**, and the **whole-plot factor** is **A**, the amount of shortening.

Next, bake six tray of cookies from each batch of dough and completely randomized the six combinations of bake temperature and tray temperature to the six trays of cookies within each batch. This is a randomized block 3×2 factorial within each batch of cookie dough and is called the **sub-plot design**.

Example 3 (Cont.): The **sub-plot block** is the **batch of cookie dough**, the **sub-plot experimental unit** is **tray of cookies**, and the **sub-plot factors** are **B**, the bake temperature, and **C,** the tray temperature.

The combination of the whole-plot and sub-plot design is called a split-plot design with CRD in the whole plots or CRSP.

By designing the experiment in this way only four batches of cookies are required instead of 12, and there are replicate whole plot and sub-plot experimental units from which to estimate the variance of the two experimental error terms.

Note that there is **less power** for **testing of shortening effect** than there would be with 12 batch completely randomized design, but due to **blocking** by batch, there is actually **more power** for detecting the baking temperature, tray temperature and interaction effects. For this reason **split-plot experiments** are often called **super efficient**.

The **optBlock** function in R package **AlgDesign** (algorithmic Experimental Design), described in chapter 4, can be used to create a randomized plan for a split-plot experiment.

```
> library(AlgDesign)

> sp = expand.grid(trayT= factor(c("Room T", "Hot")),
                   bakeT=factor(c("Low", "Mid", "High")))

> wp = data.frame(short = factor(c("100%", "80%")))
> wp = rbind(wp,wp)


> split_plot = optBlock(~short*(trayT+bakeT+trayT:bakeT),
                        withinData =sp, blocksizes =rep(6,4),
                        wholeBlockData =wp)

> split_plot$Block
```

```
> split_plot$Block

$B1
    short  trayT bakeT
1    100% Room T   Low
1.1  100%    Hot   Low
1.2  100% Room T   Mid
1.3  100%    Hot   Mid
1.4  100% Room T  High
1.5  100%    Hot  High

$B2
    short  trayT bakeT
2     80% Room T   Low
2.1   80%    Hot   Low
2.2   80% Room T   Mid
2.3   80%    Hot   Mid
2.4   80% Room T  High
2.5   80%    Hot  High

$B3
    short  trayT bakeT
3    100% Room T   Low
3.1  100%    Hot   Low
3.2  100% Room T   Mid
3.3  100%    Hot   Mid
3.4  100% Room T  High
3.5  100%    Hot  High

$B4
    short  trayT bakeT
4     80% Room T   Low
4.1   80%    Hot   Low
4.2   80% Room T   Mid
4.3   80%    Hot   Mid
4.4   80% Room T  High
4.5   80%    Hot  High
```

The data from the cookie baking experiment stored in the data frame **splitPdes** in package **daewr**.

R Codes to Obtain Data:

```
> library(daewr)
> data(splitPdes)
> y = splitPdes$y
> Library(GAD)
> Batch = as.random(splitPdes$batch)
> Short = as.fixed(splitPdes$short)
> BakeT = as.fixed(splitPdes$bakeT)
> TrayT = as.fixed(splitPdes$trayT)
> data = data.frame(y, Batch, Short, BakeT)
> data
```

	y	Batch	Short	BakeT	TrayT
1	2.11	1	100%	low	RoomT
2	2.28	1	100%	low	Hot
3	2.08	1	100%	mid	RoomT
4	2.45	1	100%	mid	Hot
5	2.46	1	100%	high	RoomT
6	2.37	1	100%	high	Hot
7	1.18	1	80%	low	RoomT
8	1.77	1	80%	low	Hot
9	1.33	1	80%	mid	RoomT
10	2.09	1	80%	mid	Hot
11	1.33	1	80%	high	RoomT
12	1.74	1	80%	high	Hot
13	2.01	2	100%	low	RoomT
14	2.14	2	100%	low	Hot
15	2.07	2	100%	mid	RoomT
16	2.26	2	100%	mid	Hot
17	2.19	2	100%	high	RoomT
18	2.34	2	100%	high	Hot
19	0.94	2	80%	low	RoomT
20	1.34	2	80%	low	Hot
21	1.23	2	80%	mid	RoomT
22	1.72	2	80%	mid	Hot
23	1.28	2	80%	high	RoomT
24	1.72	2	80%	high	Hot

Statistical Model: It is easy to write the model for a split-plot design with a completely randomized design in the whole plots by following the same two-step procedure used to create the design.

The first step was to create a completely randomized design in the whole-plots by assigning the levels of the whole-plot factor at random to the whole plots. The model for this can be represented by

$$y_{ij} = \mu + \alpha_i + w_{j(i)}$$

Shortening Effect

Shortening Effect in Batch

The second step in creating the design was to split the whole plots and assign each sub-plot at random to one of the levels of the subplot factor (or combination of levels of the sub-plot factors if there are more than one).

We should add the **effect** of **sub-plot factors** and the **interactions** between the **whole-plot** and **sub-plot factors** to the model.

Statistical Model: It is easy to write the model for a split-plot design with a completely randomized design in the whole plots by following the same two-step procedure used to create the design.

Diameter of Baked Cookie

Shortening Effect in Batch

$$y_{ijkl} = \mu + \alpha_i + w_{k(i)}$$

Tray Temp. Effect

Shortening Effect

$$+\beta_j + (\alpha\beta)_{ij} + \gamma_l$$

Bake Temp. and Tray Temp. Interaction Effect

Bake Temp. Effect

$$+(\alpha\gamma)_{il} + (\beta\gamma)_{jl}$$

Shortening and Bake Temp. Interaction Effect

$$+(\alpha\beta\gamma)_{ijl}$$

Shortening, Bake Temp. and Tray Temp. Interaction Effect

Shortening and Tray Temp. Interaction Effect

$$+\epsilon_{ijkl}$$

Error Term

With balanced data, the analysis can be performed using either the analysis of variance or the REML method.

However, if the R **aov** or **anova(model)** are used to perform the analysis of variance, it will not produce the correct F-test for the whole-plot factor.

If the terms in the model are specified as fixed or random, the **gad** function in the R package **GAD** (General ANOVA Design)can produce an ANOVA table with the correct F-test, given an objective produced by **aov** or **lm**.

```
> install.packages("GAD")
> library(GAD)
> model = aov(y~Short+Short%in%Batch+
                BakeT+ TrayT+ BakeT*TrayT+
                Short*BakeT+Short*TrayT+Short*BakeT*TrayT)
> gad(model)
```

R Codes to Obtain Correct ANOVA Table:

```
> model = aov(y~Short+Short%in%Batch+
        BakeT+ TrayT+ BakeT*TrayT+
        Short*BakeT+Short*TrayT+Short*BakeT*TrayT, data=data)
> gad(model)
```

```
Analysis of Variance Table

Response: y
                 Df Sum Sq Mean Sq F value    Pr(>F)
Short             1 3.4428  3.4428 41.0738  0.023492 *
BakeT             2 0.2053  0.1026 10.4416  0.003559 **
TrayT             1 0.6700  0.6700 68.1533 8.931e-06 ***
Short:Batch       2 0.1676  0.0838  8.5263  0.006902 **
BakeT:TrayT       2 0.0510  0.0255  2.5956  0.123608
Short:BakeT       2 0.0547  0.0274  2.7821  0.109490
Short:TrayT       1 0.1962  0.1962 19.9580  0.001202 **
Short:BakeT:TrayT 2 0.0008  0.0004  0.0424  0.958673
Residual         10 0.0983  0.0098
```

> gad(model)

```
Analysis of Variance Table

Response: y
                  Df Sum Sq Mean Sq F value    Pr(>F)
Short              1 3.4428  3.4428 41.0738  0.023492 *
BakeT              2 0.2053  0.1026 10.4416  0.003559 **
TrayT              1 0.6700  0.6700 68.1533 8.931e-06 ***
Short:Batch        2 0.1676  0.0838  8.5263  0.006902 **
BakeT:TrayT        2 0.0510  0.0255  2.5956  0.123608
Short:BakeT        2 0.0547  0.0274  2.7821  0.109490
Short:TrayT        1 0.1962  0.1962 19.9580  0.001202 **
Short:BakeT:TrayT  2 0.0008  0.0004  0.0424  0.958673
Residual          10 0.0983  0.0098
```

Note that, in the ANOVA table, it can be seen that the F-value, for example, for Short was created by dividing the mean square for Short by the mean square for batch within shortening (Short:Batch) rather than the error mean square.

The **lmer** function in the **lme4** package can also be used for REML analysis of split-plot experiments.

R Codes to Obtain Correct ANOVA Table Using REML Estimates:

```
> library(lme4)
> model2 = lmer(y~1+short+bakeT+trayT+bakeT:trayT+
                short:bakeT+short:trayT+short:bakeT:trayT+
                (1|short:batch), data = splitPdes)
> anova(model2)
```

```
Analysis of Variance Table
                    Df  Sum Sq Mean Sq F value
short                1 0.40379 0.40379 41.0738
bakeT                2 0.20530 0.10265 10.4416
trayT                1 0.67000 0.67000 68.1533
bakeT:trayT          2 0.05103 0.02552  2.5956
short:bakeT          2 0.05470 0.02735  2.7821
short:trayT          1 0.19620 0.19620 19.9580
short:bakeT:trayT    2 0.00083 0.00042  0.0424
```

The F-values above are seen identical to those produced by the **gad** function. Although the p-values are not given, they can be easily computed. For example, the p-value for the shortening effect can be calculated in R with command 1-pf(41.0738,1,2).

R Codes to Obtain REML Estimates:

> summary(model2)

Part of output is:

```
Linear mixed model fit by REML ['lmerMod']
Formula:
y ~ 1 + short + bakeT + trayT + bakeT:trayT + short:bakeT + short:trayT +
    short:bakeT:trayT + (1 | short:batch)
   Data: splitPdes

REML criterion at convergence: -8.8

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.2707 -0.4315  0.0000  0.4315  1.2707

Random effects:
 Groups      Name        Variance Std.Dev.
 short:batch (Intercept) 0.012332 0.11105
 Residual                0.009831 0.09915
Number of obs: 24, groups:  short:batch, 4

Fixed effects:
                              Estimate Std. Error t value
(Intercept)                  2.355e+00  1.053e-01  22.372
short80%                    -6.250e-01  1.489e-01  -4.198
bakeTlow                    -1.450e-01  9.915e-02  -1.462
bakeTmid                    -3.481e-16  9.915e-02   0.000
```

What Have We Learned?

We discussed nested and split-plot design in this chapter.